MINMAXexp09.ipynb - Colab https://colab.research.google.com/drive/1FLcqjWCjs...

```
import math

board = [ '' for _ in range(9)]
def print_board():
  for i in range(3):
    print(board[3*i],'|',board[3*i+1],'|',board[3*i+2])
    if i<2:
      print('--------') # Changed to `--------` for consistency


def is_winner(b,player):
  wins=[
      (0,1,2),(3,4,5),(6,7,8),
      (0,3,6),(1,4,7),(2,5,8),
      (0,4,8),(2,4,6)
  ]
  for a,b_,c in wins:
    if b[a]==b[b_]==b[c]==player:
      return True
  return False

def is_board_full(b):
  return '' not in b # Fixed spacing

def minmax(b,depth,is_maximizing):
  if is_winner(b,'X'):
    return 10-depth
  if is_winner(b,'O'):
    return depth-10
  if is_board_full(b):
    return 0

  if is_maximizing: # Fixed typo: is_maximining -> is_maximizing
    best_score=-math.inf
    for i in range(9):
      if b[i]=='':
        b[i]='X'
        score=minmax(b,depth+1,False)
        b[i]='' # backtrack
        best_score=max(score,best_score)
    return best_score # Moved return outside the loop
  else:
    best_score=math.inf
    for i in range(9):
      if b[i]=='':
        b[i]='O'
        score=minmax(b,depth+1,True)
        b[i]='' # backtrack
        best_score=min(score,best_score)
    return best_score # Moved return outside the loop
```

```
def find_best_move(): # Renamed to avoid confusion with best_score
```



```
def find_best_move(): # Renamed to avoid confusion with best_score
  best_score=-math.inf
  move=-1
  for i in range(9):
    if board[i]=='':
      board[i]='X'
      score=minmax(board,0,False)
      board[i]='' # backtrack
      if score>best_score:
        best_score=score
        move=i # Fixed typo: id -> i
  return move


def play_game():
  print("board position are numbered 0 to 8:")
  print("0|1|2")
  print("--------") # Changed to `--------` for consistency
  print("3|4|5")
  print("--------") # Changed to `--------` for consistency
  print("6|7|8")
  print()

  while True: # Added colon
    # AI's turn
    ai_move = find_best_move() # Called the renamed function
    if ai_move == -1:
      print("Game over! It's a tie.") # Corrected message
      break

    board[ai_move] = 'X'
    print("AI (X) plays move:", ai_move)
    print_board()
    if is_winner(board,'X'):
      print("AI (X) wins!")
      break

    if is_board_full(board):
      print("Game over! It's a tie.") # Corrected message
      break

    # Human's turn
    while True:
      try:
        human_move = int(input("Enter your move (0-8):"))
        if 0 <= human_move <= 8 and board[human_move] == '':
```

```
        board[human_move] = 'O' # Assuming human is 'O'
        break
      else:
        print("Invalid move. Try again.")
    except ValueError:
      print("Please enter a valid number.")

  print_board()
if is_winner(board,'O'):
```



```
  if is_winner(board,'O'):
    print("You win!")
    break

  if is_board_full(board):
    print("Game over! It's a tie.") # Corrected message
    break

if __name__ == "__main__": # Corrected indentation and __name__
  play_game()
```

```
board position are numbered 0 to 8:
0|1|2
--------
3|4|5
--------
6|7|8

AI (X) plays move: 0
X | |
--------
 | |
--------
 | |
```

Enter your move (0-8):

Start coding or generate with AI.